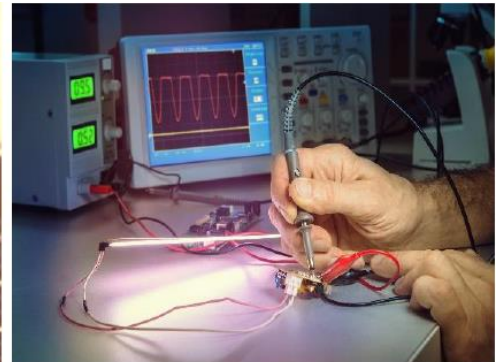
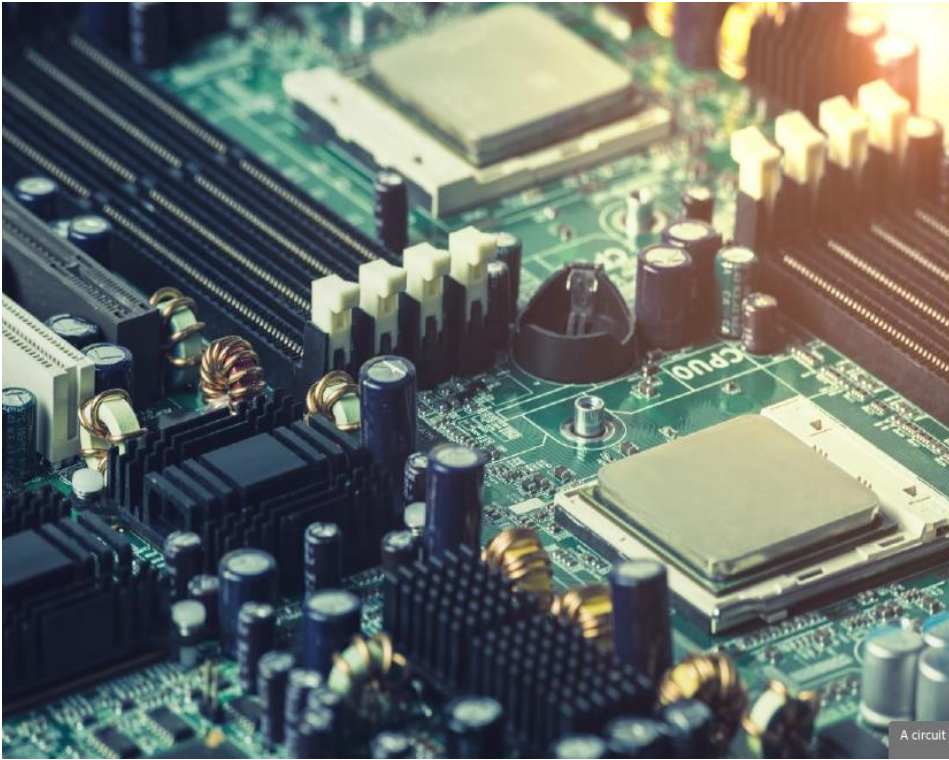


Model 670C

16-Bit Arbitrary Waveform Generator



A circuit board

Features

- 2 or 4 Analog Channels
- 600 MS/s (1.2 GS/s with x2 interpolation)
- 16-bit Vertical Resolution
- Up to 180 MHz Bandwidth
- Up to 12V_{p-p} into 50Ω load
- Up to 512Mpts Waveform Memory per Channel
- 8 Digital Channels in synchronous with analog Generation
- Simple Rider™ UI: designed for touch AWG/AFG user interfaces



Model 670C
4 Channel Arbitrary Waveform
Generator

Applications

- Automotive
- IoT and Ind 4.0 perfect RF Modulator
- Research
- Aerospace & Defense
- Semiconductors

INTRODUCTION

The Model 670C is the cost-effective and powerful 2 or 4 channel Arbitrary Function Generator (AFG) and 2 or 4 channel Arbitrary Waveform Generator (AWG) with advanced sequencer functionality.

180 MHz bandwidth, up to 12Vpp output range and up to 512 Msample memory depth; the 670C is the ideal Arbitrary Waveform Generator choice for automotive, IoT, and medical applications.

The 8 channel digital option combined with 2 or 4 analog channels makes the Model 670C a full featured mixed signal generator. It's now possible to generate 2 or 4 analog signals fully synchronized with 8 digital lines (LVTTTL or LVDS standards).

SPECIFICATIONS

General Specifications

	670C-2C	670C-4C / 670C-4C-LF
Number of Channels		
Analog out	2	4
Digital out	0/8 – optional	0/8 – optional
Marker out	1	1
Operating Mode	AFG Mode True Arb Mode	
	670C	670C-4C-LF
Amplitude		
Range (50Ω into 50Ω) ¹	0 to 6Vp-p (12Vp-p optional)	0 to 12Vp-p
Accuracy (1kHz sine wave, 0V offset, >5mVp-p amplitude, 50Ω load) (guaranteed)	±(1% of setting [Vp-p] + 5mV)	
Resolution	<0.5mVp-p or 5 digits	
Output Impedance	Single-ended: 50Ω, Low Impedance: 0Ω	
DC		
Amplitude range (50Ω into 50Ω) ¹	-3V to 3V (-6V to 6V optional)	-6V to 6V
Amplitude accuracy (guaranteed)	±(1% of setting + 10mV)	
Output Attenuator	0dB or 20dB selectable	

AFG Mode Specifications

Output Channels	
Connectors	BNC on front panel
Output Type	Single-ended
Output Impedance	50Ω or 0Ω (low impedance) programmable
General Specifications	
Operating Mode	DDS mode
Standard Waveforms	Sine, Square, Pulse, Ramp, more (Noise, DC, Sin(x)/x, Gaussian, Lorentz, Exponential Rise, Exponential Decay, Haversine
Run Modes	Continuous, modulation, sweep, burst
Arbitrary Waveforms	Vertical resolution: 16-bit
Internal Trigger Timer	Waveform length: 16,384 points
Range	13.4ns to 100s
Resolution	104ps
Accuracy	±(0.1% setting + 5ps)

	670C	670C-4C-LF
Sine Waves		
Frequency Range (50Ω into 50Ω)	1 μHz to ≤ 150 MHz: 6 V _{p-p} >150 MHz to ≤ 180 MHz: 5 V _{p-p} HV option: 1 μHz to ≤ 50 MHz: 12 V _{p-p} >50 MHz to ≤ 60 MHz: 10 V _{p-p} >60 MHz to ≤ 100 MHz: 8 V _{p-p} >100 MHz to ≤ 150 MHz: 6 V _{p-p} >150 MHz to ≤ 180 MHz: 5 V _{p-p}	1 μHz to ≤ 50 MHz: 12 V _{p-p} >50 MHz to ≤ 60 MHz: 10 V _{p-p} >60 MHz to ≤ 100 MHz: 8 V _{p-p}
Max Frequency Value	180 MHz	100 MHz
Flatness (1V _{p-p} , relative to 1 kHz)	DC to 180 MHz: ±0.5dB	DC to 100 MHz: ±0.5dB
Harmonic Distortion (1V _{p-p})	1 μHz to ≤ 20 kHz: <-75dBc >20 kHz to ≤ 1 MHz: <-70dBc >1 MHz to ≤ 10 MHz: <-65dBc >10 MHz to ≤ 50 MHz: <-55dBc >50 MHz to ≤ 120 MHz: <-45dBc >120 MHz to ≤ 180 MHz: <-40dBc	1 μHz to ≤ 20 kHz: <-75dBc >20 kHz to ≤ 1 MHz: <-70dBc >1 MHz to ≤ 10 MHz: <-65dBc >10 MHz to ≤ 50 MHz: <-55dBc >50 MHz to ≤ 100 MHz: <-45dBc
Total Harmonic Distortion (1V _{p-p})	10 Hz to 20 kHz: <0.04%	
Spurious (1V _{p-p}) (excluding f _{Sa} -f _{Out} , f _{Sa} -2*f _{Out})	1 μHz to ≤ 10 MHz: <-80dBc >10 MHz to ≤ 180 MHz: <-80dBc + 6dBc/octave	1 μHz to ≤ 10 MHz: <-80dBc >10 MHz to ≤ 100 MHz: <-80dBc + 6dBc/octave
Phase Noise (1V _{p-p} , 10kHz offset)	10 MHz: < -127dBc/Hz typ 100 MHz: < -115dBc/Hz typ	
Square Waves		
Frequency range	1 μHz to 80 MHz: 6V _{p-p} HV option: 1 μHz to ≤ 30 MHz: 12V _{p-p} >30 MHz to ≤ 50 MHz: 11V _{p-p} >50 MHz to ≤ 70 MHz: 10V _{p-p} >70 MHz to ≤ 80 MHz: 9V _{p-p}	1 μHz to ≤ 30 MHz: 12V _{p-p} >30 MHz to ≤ 50 MHz: 11V _{p-p}
Rise/fall time	4ns	5ns
Overshoot (1V _{p-p})	<1%	<1%
Jitter (rms)	<2ps	<2ps
Pulse Waves		
Frequency Range	1 μHz to 80 MHz: 6V _{p-p} HV Option: 1 μHz to ≤ 3 MHz: 12V _{p-p} >3 MHz to ≤ 10 MHz: 11V _{p-p} >10 MHz to ≤ 70 MHz: 10V _{p-p} >70 MHz to ≤ 80 MHz: 9V _{p-p}	1 μHz to ≤ 3 MHz: 12V _{p-p} >3 MHz to ≤ 10 MHz: 11V _{p-p} >10 MHz to ≤ 50 MHz: 10V _{p-p}
Leading/trailing edge transition time	4ns to 1000s	5ns to 1000s
Pulse width	5ns to (Period – 5ns)	7ns to (Period – 7ns)
Pulse width resolution	20ps or 15 digits	
Transition time resolution	2ps or 15 digits	
Pulse duty	0% to 100% 14 digits (limitations of pulse width apply)	
Overshoot (1V _{p-p})	<1%	
Jitter (rms, with rise and fall time ≥4ns)	<2ps	
Double Pulse Waves		
Frequency Range	1 μHz to ≤ 3 MHz: 12V _{p-p} >3 MHz to ≤ 50 MHz: 6V _{p-p} where V _{p-p} = V _{p-p1} + V _{p-p2} HV option: 1 μHz to ≤ 3 MHz: 24V _{p-p} >3 MHz to ≤ 10 MHz: 11V _{p-p} >10 MHz to ≤ 50 MHz: 10V _{p-p}	1 μHz to ≤ 3 MHz: 24V _{p-p} >3 MHz to ≤ 10 MHz: 11V _{p-p} >10 MHz to ≤ 50 MHz: 10V _{p-p} where V _{p-p} = V _{p-p1} + V _{p-p2}

	where $V_{p-p} = V_{p-p1} + V_{p-p2} $	
Other Pulse Parameters	Same as pulse waves	
Ramp Waves		
Frequency Range	1 μ Hz to 5 MHz	
Linearity (<10 kHz, 1Vp-p, 100%)	$\leq 0.1\%$	
Symmetry	0% to 100%	
Other Waves		
Frequency Range		
Exponential Rise, Exponential Decay	1 μ Hz to 5 MHz	
Sin(x)/x, Gaussian, Lorentz, Haversine	1 μ Hz to 10 MHz	
Additive Noise		
Bandwidth (-3dB)	>200 MHz	>100 MHz
Level	0V to 6V – carrier max value $[V_{pk}]$	
Resolution	1 mV	
Arbitrary		
Number of Samples	2 to 16,384	
Jitter (rms)	<2ps	
Frequency Range	1 μ Hz to ≤ 80 MHz	1 μ Hz to ≤ 50 MHz
Analog Bandwidth (-3 dB)	87.5 MHz	70 MHz
Rise/fall time	4ns	5ns
Frequency Resolution		
Sine, Square, Pulse, Arbitrary, Sin(x)/s	1 μ Hz or 15 digits	
Gaussian, Lorentz, Exponential Rise, Exponential Decay, Haversine	1 μ Hz or 14 digits	
Frequency Accuracy		
Non-ARB	$\pm 2.0 \times 10^{-6}$ of setting	
ARB	$\pm 2.0 \times 10^{-6}$ of setting $\pm 1 \mu$ Hz	

Modulations

Amplitude Modulation (AM)		
Carrier waveforms	Standard waveforms (except Pulse, DC and Noise), ARB	
Modulation source	Internal	
Internal modulating waveforms	Sine, Square, Ramp, Noise, ARB	
Modulating frequency	500 μHz to 48 MHz	
Depth	0.00% to 120.00%	
Frequency Modulation (FM)		
Carrier waveforms	Standard waveforms (except Pulse, DC and Noise), ARB	
Modulation source	Internal	
Internal modulating waveforms	Sine, Square, Ramp, Noise, ARB	
Modulating frequency	500 μHz to 48 MHz	
Peak Deviation	DC to 180 MHz	DC to 100 MHz
Phase Modulation (PM)		
Carrier waveforms	Standard waveforms (except Pulse, DC and Noise), ARB	
Modulation source	Internal	
Internal modulating waveforms	Sine, Square, Ramp, Noise, ARB	
Modulating frequency	500 μHz to 48 MHz	
Phase deviation range	0° to 360°	
Frequency Shift Keying (FSK)		
Carrier waveforms	Standard waveforms (except Pulse, DC and Noise), ARB	
Modulation source	Internal	
Internal modulating waveforms	Square	
Key rate	500 μHz to 48 MHz	
Number of keys	2	

Hop frequency	1 μHz to 180 MHz	1 μHz to 100 MHz
Phase Shift Keying (PSK)		
Carrier waveforms	Standard waveforms (except Pulse, DC and Noise), ARB	
Modulation source	Internal	
Internal modulating waveforms	Square	
Key rate	500 μHz to 48 MHz	
Hop phase	0° to +360°	
Number of keys	2	
Pulse Width Modulation		
Carrier waveforms	Pulse	
Modulation source	Internal	
Internal modulating waveforms	Sine, Square, Ramp, Noise, ARB	
Modulating frequency	500 μHz to 48 MHz	
Deviation range	0% to 50% of pulse period	
Sweep		
Type	Linear, Logarithmic, Staircase, and user defined	
Waveforms	Standard waveforms (except Pulse, DC and Noise), ARB	
Sweep time	40ns to 2000s	
Hold/return times	0 to (2000s – 40ns)	
Sweep/hold/return time resolution	20ns or 12digits	
Total sweep time accuracy	≤0.4%	
Trigger source	Internal / External / Manual	
Start/stop frequency range	Sine: 1 μHz to 180 MHz Square: 1 μHz to 80 MHz	Sine: 1 μHz to 100 MHz Square: 1 μHz to 50 MHz
Burst		
Waveforms	Standard waveforms (except DC and Noise), ARB	
Type	Triggered or Gated	
Burst count	1 to 4,294,967,295 cycles or Infinite	

True Arb Mode Specifications

Output Channels		
Connectors	BNC on front panel	
Output type	Single-ended DC coupled	
Output Impedance	50Ω or 0Ω (low impedance)	
	670C	670C-4C-LF
General specifications		
Operating mode	Variable clock (True Arbitrary)	
Run Modes	Continuous, Triggered Continuous, Single/Burst, Stepped, Advanced	
Vertical Resolution	16 bit	
Waveform length	16 to 2M samples per channel (up to 512 samples optional)	16 to 2M samples per channel (up to 128M samples optional)
Waveform Granularity	1 if the entry length is >384 samples 8 if entry length is ≥16 and ≤384 samples	
Sequence length	1 to 16,384	
Sequence repeat counter	1 to 4,294,967,295 or infinite	
Timer		
Range	23.52ns to 7s	
Resolution	±1 sampling clock period	
Analog channel to channels skew		
Range	0 to 6.59 us (depending on internal sampling rate)	
Resolution	Channel 1/2 to Channel 3/4: ≤ 5ps, Channel 1/3 to Channel 2/4: 1 DAC sampling period	
Accuracy	±(1% of setting + 20ps)	
Initial Skew	<200 ps	
Calculated Bandwidth (0.35 / rise or fall time) ²	≥160 MHz	≥100 MHz
Harmonic Distortion (Sine wave 32pts, 1V _{p-p})	< -62dBc (@ 600MS/s, 18.75 MHz)	
Spurious (Sine wave 32 pts, 1V _{p-p})	< -80dBc (@ 600MS/s, 18.75 MHz)	
SFDR (Sine wave 32 pts, 1V _{p-p} , including harmonics)	< -62dBc (@ 600MS/s, 18.75 MHz)	
Rise/fall time (1V _{p-p} single-ended 10% to 90%) ²	≤2.2ns	≤3.5ns
Overshoot (1V _{p-p} single-ended) ²	< 2%	
Timing & Clock		
Sampling Rate		
Range	1 S/s to 600 MS/s (1 S/s to 1.2 GS/s with x2 interpolation)	
Resolution	16 Hz	
Accuracy	±2.0ppm	
Random jitter on clock pattern (rms)	<2ps	
Digital outputs		
Optional		
Output channels		
Connectors	Mini-SAS HD connector on rear panel (non-standard pin-out)	
Number of connectors	1	
Number of outputs	8 bits	
Output impedance	100Ω differential	
Output type	LVDS	
Rise/fall time (10% to 90%)	<1ns	
Jitter (rms)	20ps	
Maximum update rate	600 Mbps	
Memory depth	2MSamples per digital channel (up to 512MSamples optional)	

8 vit LVDS to LVTTTL Converter Probe Optional AT-DTTL8	
Output connector	20 position 2.54 mm 2 Row IDC Header
Output type	LVTTTL
Output impedance	50 Ω nominal
Output voltage	0.8V to 3.8V programmable
Maximum update rate	125Mbps@0.8V and 400Mbps@3.6V
Dimensions	W 52mm – H 22mm – D 76mm
Input connector	Proprietary standard
Cable length	1 meter
Cable type	Proprietary standard
Proprietary Mini SAS HD to SMA cable Optional	
Output connector	SMA
Output type	LVDS
Number of SMA	16 (8 bits)
Cable type	Proprietary standard
Cable length	1 meter

Auxiliary Input & Output Characteristics

	670C-2C	670C-4C-LF, 670C-4C
Marker output		
Connector type	BNC on front panel	BNC on rear panel
Number of connectors	1	
Output impedance	50 Ω	
Output level (into 50 Ω)		
Amplitude	1V to 2.5V	
Resolution	10mV	
Accuracy	±(2% setting + 10mV)	
Rise/fall time (10% to 90%, 2.5V _{p-p})	<700ps	
Jitter (rms)	20ps	
Marker out to analog channel skew		
Range	True Arb Mode: 0 to 3μs AFG Mode: 0 to 14s in Continuous Mode 0 to 3μs in Triggered Mode	
Resolution	True Arb Mode: 78ps, AFG Mode: 39ps	
Accuracy	±(1% of setting + 140 ps)	
Initial Skew	< 1 ns	
Trigger/Gate Input		
Connector	BNC on front panel	BNC on rear panel
Input impedance	50Ω / 1kΩ programmable	
Slope/polarity	Positive or negative or both	
Input damage level	<-15V or >+15V	
Threshold control level	-10V to 10V	
Resolution	10mv	
Threshold control accuracy	±(10% of setting + 0.2V)	
Input voltage swing	0.5V _{p-p} minimum	
Minimum pulse width (1V _{p-p})	3ns	
Initial trigger/gate delay to Analog Output	AFG mode: <400 ns (<460 ns in triggered sweep mode) True Arb mode: <131*DAC sampling period + 22.5 ns (<143*DAC sampling period+22.5 ns with 2x interpolation)	
Trigger In to output jitter	AFG mode: <45ps	

Maximum frequency	True Arb mode: 0.29*DAC sampling period
	AFG mode: 55 MTps on Rising/Falling Edge, 80 MTps on Both Edges
	True Arb mode: 42.5 MTps on Rising/Falling Edge, 42.5 MTps on Both Edges
	where MTps = Mega Transitions per second
Reference clock input	
Connector type	SMA on rear panel
Input impedance	50Ω, AC coupled
Input voltage range	-4 dBm to 11dBm sine or square wave (rise time T_{10-90} <1ns and duty cycle from 40% to 60%)
Damage level	+14dBm
Frequency range	5 MHz to 100 MHz
Reference clock output	
Connector type	SMA on rear panel
Output impedance	50Ω, AC coupled
Frequency	10 MHz
Accuracy	$\pm 2.0 \times 10^{-6}$
Aging	$\pm 1.0 \times 10^{-6}$ /year
Amplitude	1.65V
Jitter (rms)	<20ps

Power

Source Voltage & Frequency	100 to 240VAC $\pm 10\%$ @ 45 Hz to 66 Hz
Max. power consumption	100W

Environmental Characteristics

Temperature (operating)	+5°C to +40°C (+41°F to 104°F)
Temperature (non-operating)	-20°C to +60°C (-4°F to 140°F)
Humidity (operating)	5% to 80% relative humidity with a maximum wet bulb temperature of 29°C at or below +40°C, (upper limit de-rates to 20.6% relative humidity at +40°C). Non-condensing.
Humidity (non-operating)	5% to 95% relative humidity with a maximum wet bulb temperature of 40°C at or below +60°C, upper limit de-rates to 29.8% relative humidity at +60°C. Non-condensing.
Altitude (operating)	3,000 meters (9,842 feet) maximum at or below 25°C
Altitude (non-operating)	12,000 meters (39,370 feet) maximum

EMC & Safety

Compliance	CE compliant
Safety	EN61010-1
Main Standards	EN 61326-1:2013 – Electrical equipment for measurement, control and laboratory use – EMC requirements – Part 1: General requirements
Immunity	EN 61326-1:2013

System Specifications

	670C-2C	670C-4C, 670C-4C-LF
Display	7", 1024x600, capacitive touch LCD	

Operative System	Windows 10	
External Dimensions	W 362 mm – H 143 mm – D 258 mm (3U 10" rackmount)	
Weight	6.25 kg	
Front panel connectors	CH1, CH2 OUTPUT (BNC) MARKER OUT (BNC) TRIGGER IN (BNC)	CH1, CH2 OUTPUT (BNC) CH3, CH4 OUTPUT (BNC)
Rear panel connectors	REF CLK IN (SMA) REF CLK OUT (SMA) External Monitor ports DIGITAL POD A[7..0] 1 USB 2.0 ports or more Ethernet port (10/100/1000BaseT Ethernet, RJ45 port) 2 PS/2 keyboard and mouse ports	REF CLK IN (SMA) REF CLK OUT (SMA) MARKER OUT (BNC) TRIGGER IN (BNC) External Monitor ports DIGITAL POD A[7..0] 1 USB 2.0 ports or more Ethernet port (10/100/1000BaseT Ethernet, RJ45 port) 2 PS/2 keyboard and mouse ports
Hard disk	240 GB SSD or better	
Processor	Intel® Celeron J1900, 2 GHz (or better)	
Processor Memory	4 GB or better	

TABLE OF AVAILABLE MODELS

Item	Description
670C-2C-2M	2ch 600Ms/s AWG 2Ms memory - 180MHz AFG
670C-2C-64M	2ch 600Ms/s AWG 64Ms memory - 180MHz AFG
670C-2C-256M	2ch 600Ms/s AWG 256Ms memory - 180MHz AFG
670C-2C-512M	2ch 600Ms/s AWG 512Ms memory - 180MHz AFG
670C-4C-2M	4ch 600Ms/s AWG 2Ms memory - 180MHz AFG
670C-4C-64M	4ch 600Ms/s AWG 64Ms memory - 180MHz AFG
670C-4C-256M	4ch 600Ms/s AWG 256Ms memory - 180MHz AFG
670C-4C-512M	4ch 600Ms/s AWG 512Ms memory - 180MHz AFG
670C-4C-2M-LF	4ch 600Ms/s AWG 2Ms memory – 100MHz AFG
670C-4C-128M-LF	4ch 600Ms/s AWG 128Ms memory – 100MHz AFG

TABLE OF AVAILABLE OPTIONS & ACCESSORIES

Item	Description
Options	
670C-2C-HV	High voltage output (12Vpp on 50ohm)
670C-4C-HV	High voltage output (12Vpp on 50ohm)
Accessories	
AT-DTTL8	LVDS to LVTTTL digital adapter probe
AT-LVDS-SMA8	LVDS to SMA digital adapter cable
RIDER-C-RACK	Rackmount kit for Rider C series
GPIB / USB-TMC	GPIB and USBTMC Ports for Remote Control
SSD-250	Additional 250GB Solid State Disk for RIDER series

SSD-500	Additional 500GB Solid State Disk for RIDER series
SSD-1000	Additional 1TB Solid State Disk for RIDER series
8DIG	8 channel digital license

DOCUMENT HISTORY

Version	Date	Author	Notes
V1.0	1/13/25	At	New datasheet
V1.2	6/10/25	At	LF option

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